

TSETSEN UUL, Mongolia

WMO: 442240

Lat: 48.748N

Lon: 96.004E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.1	-43.6	-52.3	0.3	-48.7	-47.3	0.5	-43.3	17.2	9.6	16.1	6.9	0.3	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	77.0	56.2	73.8	54.7	70.8	53.4	59.0	71.3	57.0	69.0	55.2	67.4	5.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	83.5	64.1	52.1	75.7	61.7	50.2	70.3	60.7	29.2	71.3	27.6	68.9	26.3	67.4	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.8	14.1	-51.6	82.6	7.1	3.1	-56.7	84.8	-60.8	86.6	-64.8	88.4	-69.9	90.7		
(5)			-51.4	61.0	7.6	1.2	-56.8	61.9	-61.3	62.6	-65.5	63.2	-71.0	64.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	-20.0	-13.1	5.6	30.3	41.3	53.3	57.4	52.8	41.2	25.6	4.2	-12.7		
	DBStd	29.09	12.62	13.18	15.25	11.46	8.14	6.61	5.43	6.42	7.94	10.79	13.33	12.21		
	HDD50	10634	2169	1767	1375	591	286	43	6	45	280	755	1373	1944		
	HDD65	15574	2634	2187	1840	1040	735	353	239	380	714	1220	1823	2409		
	CDD50	544	0	0	0	1	16	141	237	132	17	0	0	0		
	CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0		
	CDH74	259	0	0	0	0	2	49	148	59	1	0	0	0		
	CDH80	24	0	0	0	0	0	1	19	4	0	0	0	0		
	WSAvg	4.3	2.0	2.7	4.3	5.7	6.6	4.9	4.4	4.3	4.9	4.7	4.5	2.6		
	PrecAvg	8.7	0.1	0.1	0.2	0.3	0.6	1.3	2.4	2.0	0.7	0.3	0.3	0.2		
(15) Precipitation	PrecMax	13.1	0.5	0.4	0.5	0.9	1.7	2.4	5.4	3.7	2.3	0.7	0.6	0.5		
	PrecMin	5.5	0.0	0.0	0.0	0.0	0.1	0.3	0.4	0.5	0.1	0.0	0.1	0.0		
	PrecStd	2.0	0.1	0.1	0.1	0.2	0.4	0.5	1.3	0.8	0.5	0.2	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	31.5	44.0	61.0	71.6	78.2	83.2	80.0	71.2	55.0	44.2	27.5		
		MCWB	18.2	25.5	33.1	45.2	50.5	56.2	61.2	56.4	52.8	40.8	35.0	21.2		
	2%	DB	14.0	22.2	39.2	56.3	66.6	75.1	78.1	75.1	65.9	50.5	38.4	21.5		
		MCWB	10.9	17.2	30.5	42.9	48.5	54.2	56.9	54.6	50.0	38.6	30.5	17.5		
	5%	DB	9.6	16.0	34.9	52.3	62.7	71.6	74.5	71.3	61.9	47.1	31.8	14.8		
		MCWB	6.9	12.7	27.8	39.9	46.2	52.8	55.8	53.5	47.7	36.3	26.2	11.7		
	10%	DB	3.4	10.6	31.0	48.4	58.9	68.6	71.3	67.6	58.1	43.6	25.8	9.7		
		MCWB	1.6	7.9	25.2	37.4	43.8	51.7	54.7	52.2	45.4	34.1	21.5	7.9		
	0.4%	WB	18.1	25.3	34.0	46.2	53.1	59.1	64.3	60.1	55.1	42.2	35.6	21.6		
		MCDB	22.4	31.1	43.0	59.3	68.4	73.9	77.7	73.5	66.9	52.6	43.7	26.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.0	17.4	30.8	43.2	49.6	56.6	60.6	57.4	51.0	39.8	32.5	17.6		
		MCDB	13.9	21.9	38.4	55.4	64.6	70.7	71.9	69.4	64.2	50.2	39.5	21.7		
	5%	WB	7.0	12.6	28.2	40.4	47.0	54.7	58.0	55.3	48.3	37.7	28.4	12.0		
		MCDB	9.6	16.1	34.8	51.6	60.9	67.4	69.1	67.8	60.0	47.2	34.2	14.8		
	10%	WB	1.6	8.1	25.1	37.6	44.7	52.9	56.2	53.4	46.2	35.3	23.9	7.7		
		MCDB	3.5	10.6	30.9	47.6	57.6	65.6	67.8	65.3	56.5	44.0	28.2	9.8		
	5% DB	MDBR	24.8	29.8	31.3	22.1	23.5	23.7	21.6	22.8	24.7	24.8	23.6	22.7		
		MCDBR	30.1	32.5	29.5	26.0	31.4	29.4	29.2	30.0	31.4	29.3	26.2	26.2		
		MCWBR	27.5	28.8	22.6	15.7	17.9	14.5	13.7	14.7	18.2	17.8	12.8	23.2		
		MCWBR	30.3	32.3	28.6	24.9	28.3	25.5	23.4	25.3	28.9	27.9	18.3	26.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	27.7	28.8	22.2	15.7	17.3	13.6	12.6	13.5	17.2	18.2	12.4	23.4		
(40) Clear-Sky Solar Irradiance	taub		0.213	0.237	0.275	0.314	0.316	0.318	0.320	0.303	0.270	0.255	0.235	0.217		
	taud		2.343	2.290	2.258	2.296	2.332	2.409	2.437	2.457	2.504	2.472	2.393	2.326		
	Ebn at Noon		283	297	299	297	299	298	296	298	300	288	270	265		
	Edh at Noon		21	28	35	38	38	36	34	32	27	23	20	18		
(44) All-Sky Solar Radiation	RadAvg		500	845	1349	1791	2114	2160	2033	1775	1461	950	515	383		
	RadStd		37	44	47	78	110	77	107	78	67	53	49	32		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	-3.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page